NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-27/72A
18 May 1977
SUPERSEDING
MIL-T-27/72
14 April 1976

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, AUDIO FREQUENCY TF5S21ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformers described herein shall consist of this document and the latest issue of Specification MIL-T-27.

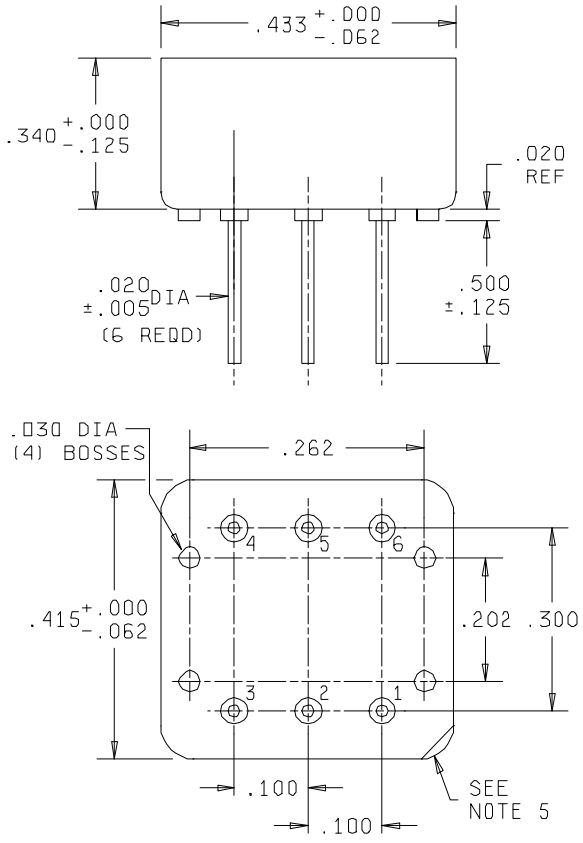
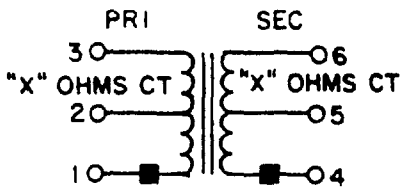


FIGURE 1. Dimensions and configuration.

FSC 5950

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INCHES	MM	INCHES	MM
005	13	202	5 13
020	51	262	6 65
030	76	300	7 62
062	1 57	340	8 64
100	2 54	415	10 54
125	3 18	433	11 00
		500	12 70



WORKING VOLTAGE 175 V
ALTITUDE 70kFT MAX
(SEE TABLE I FOR VALUE OF "X")

CIRCUIT DIAGRAMS AND MARKING

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm
3 Unless otherwise specified, tolerance is ± 005 (13 mm)
4 Marking shall be on sides and/or top of case
5 Beveled corner shall be 45° and identifies pin no 1
6 Electrical values shall be marked as specified in table I, as applicable

FIGURE 1 Dimensions and configuration - Continued

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-3) are used, they indicate the winding and the extreme terminals of the winding)

Electrical ratings See table I.
Working voltage: 175 volts.
Frequency range 300 hertz to 100 kilohertz (except 300 hertz to 40 kilohertz for M27/72-01).

TABLE I. Electrical ratings.1/

Dash No	Primary impedance ohms CT (1-3)	Secondary impedance ohms CT (4-6)	Primary DCR ohms ±25%	Primary power level at 300 Hz mW (max)	Secondary DCR ohms ±25%	Primary unbalanced DC current (max) mA
01	100 k	1,000	6,800	10	110	0.2
02	25 k	1,000	4,300	50	140	1.2
03	25 k	600	4,300	50	90	1.25
04	10 k	10 k	1,350	50	1,350	1.0
05	10 k	1,500	1,350	50	170	1.0
06	10 k	600	1,350	50	90	1 0
07	1,000	600	140	50	90	6.25
08	600	600	90	50	90	8.0
09	600	60	90	50	10	8.0

1/ Qualification testing to M27/72-01 and all tests per table VII of MIL-T-27 for M27/72-02 shall be sufficient to grant qualification approval to M27/72-01 through M27/72-09.

Design and construction
Dimensions and configuration See figure 1.
Duty cycle Continuous.
Case Encapsulated.
Material Plastic
Terminals Pin (printed circuit)
Material Nickel, tin plated
Diameter 0 020 ±0 005 inch.
Length 0 500 ±0 125 inch
Weight 3 97 grams maximum
Altitude 70,000 feet maximum
Operating temperature range -55°C to +130°C.

Terminal strength MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage
At sea level 500 volts rms.
At barometric pressure 300 volts rms

Electrical characteristics See table II
Frequency response
Reference 1 kilohertz

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TABLE II. Electrical characteristics

Dash No	Frequency response at 300 Hz to 100 kHz ±2dB			
	Zs (1-3) ohms	Z_L (4-6)	E_L volts	Primary DC current (1-3) mA
01 1/	100 k	1 k	10	0.1
02	25 k	1 k	5.0	0.6
03	25 k	600	5 0	0.63
04	10 k	10 k	2.0	0.5
05	10 k	1,500	2.0	0.5
06	10 k	600	2.0	0.5
07	1 k	600	1.0	3.12
08	600	600	0.6	4.0
09	600	60	0.6	4.0

1/ Frequency response is 300 hertz to 40 kilohertz ±2 dB

Marking location: See figure 1.

Part number M27/72-(dash number from table I).

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians
Army - EL
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - MU, WC, ME
Navy - MC, AS, SH, OS
Air Force - 19

Preparing activity
Army - EL

Agent
DLA - ES

(Project 5950-0543-6)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-T-27/72A TRANSFORMERS, AUDIO FREQUENCY TF5S21ZZ

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

ESC-FM 1068-68

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